PAPER ID-411290

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM IV) THEORY EXAMINATION 2023-24
SIGNAL SYSTEM

TIME: 3 HRS **M.MARKS: 70**

Note: **1.** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 7 = 14**

a.	Determine the fundamental period of the signal **x(t)= sin(4t-1).**	2
b.	Write the general formula for calculating Convolution in Continuous Time Domain.	2
c.	Find the Fourier Transform of continuous time signal **x(t) = δ(t + 1).**	2
d.	Calculate the Z-Transform of discrete time signal **x(n)= δ(n + 1).**	2
e.	Sketch the Sampled Signal of **x(t)=cos(2πt).**	2
f.	Sketch the continuous time signal **x(t)=u(-t+1).**	2
g.	What is Parseval's Theorem?	2

SECTION B

2. Attempt any *three* of the following: **7 x 3 = 21**

a.	Define the following terms related to system with mathematical expressions (i) Linear System (ii) Stability (iii) Causality and (iv) Dynamic system.	7
b.	A causal LTI system is described by difference equation. **y(n) = y(n-1) +y(n-2) + x(n-1)** Find the system function H(z)=Y(z)/X(z) for this system. Plot the pole-zero plot of H(z) and indicate the region of convergence.	7
c.	Sketch the signal **y(t)=e^{-t}u(t)** & find the Fourier Transform of this same signal with Magnitude & Phase Curve.	7
d.	What is ROC? Discuss any three properties of Z-Transform with mathematical expressions and example.	7
e.	Describe Natural Sampling process with graphs and mathematical equations.	7

SECTION C

3. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Define Time Invariant system. Also check whether the given system is Time Variant or Invariant system **(i) $y_1(t)=tx(t)$, (ii) $y_2(n)=2x(n)+3$, (iii) $y_3(t)=2x(-t)$.**	7
b.	Define the following with mathematical expressions and examples (i) Energy & Power Signals, (ii) Even & odd Signals.	7

4. Attempt any *one* part of the following: **7 x 1 = 7**

a.	What is Discrete Time Convolution? Find the convolution y(n)= x(n)*h(n) of the discrete time signals x(n)=h(n)=u(n).	7
b.	What is LTI System? An LTI system is described by differential equation $\frac{dy(t)}{dt}+5y(t)=3x(t)$. Calculate (i) Transfer Function, H(s)=Y(s)/X(s). (ii) Output, y(t) when x(t)=u(t).	7

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5. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Write any three properties of Fourier Transform. Find the Fourier Transform of the given signal using Properties $\mathbf{y(t)=te^{-4t}u(t)}$.	7
b.	Determine the Inverse Laplace transform of the following functions using Partial Fraction method: **(i)** $X(s) = \frac{4(s+3)}{s(s+1)(s+2)}$ **(ii)** $Y(s) = \frac{4}{(s+1)(s+2)^2}$	7

6. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Determine the Inverse Z-Transform y(n) of the following functions $Y(z) = \frac{z^2}{(z^2+3z+2)}$, given **ROC** $\mathbf{|z|>2}$. https://www.aktuonline.com	7
b.	Determine the z-transform of $\mathbf{x(n) = (1/2)^n u(n)}$ and depict the ROC and the location of poles and zeros in the z plane.	7

7. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Explain Ideal Sampling with time domain & frequency domain graphs and mathematical equations.	7
b.	What is Nyquist Rate & Nyquist Interval? Calculate the Nyquist rate & Nyquist Interval for the following continuous-time sinusoidal signals (i) $\mathbf{x_1(t)=cos(20\pi t)+cos(40\pi t)}$. **and** (ii) $\mathbf{x_2(t)=cos(200\pi t)cos(400\pi t)}$.	7

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